

Doping Stability Report -- Mn in KCoO2 at 200 C

Generated: 2026-05-30 17:26 Host: KCoO2 Dopant: Mn

Executive Summary

Mn preferentially occupies the Co layer. The formation energy difference between the two layers is +6.077 eV, strongly favouring the Co site. MD confirms this preference: on the Co site the dopant MSD plateau slope is 0.00543 Å²/ps (stable), compared to 0.00629 Å²/ps on the K site. The Co-site result is isovalent (no charge mismatch), so the formation energy is fully reliable. The K-site E_f is approximate due to a charge surplus of +2 that cannot be modelled in CHGNet.

Site	E _f (eV)	Charge	Compensation	MSD pass	Coord pass	Verdict
Co	-2.526	+0	No	FAIL	PASS	MIGRATION (favorable site but
K	+3.551	+2	Yes	FAIL	PASS	MIGRATION / UNFAVORABLE

Mn at the Co layer -- MIGRATION (favorable site but dopant relocates during MD)

Charge situation

Mn replaces Co with no change in oxidation state (isovalent substitution, charge mismatch = 0). No charge compensation is needed. The formation energy is fully reliable.

Formation Energy (thermodynamic stability)

Formation energy E _f	-2.5260 eV	PASS	< 1.0 eV to pass
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E_f = -2.526 eV. This value is strongly negative, indicating that Mn incorporation is thermodynamically very favorable at the Co site.

Molecular Dynamics Stability (200 C, 25 ps)

MSD slope (late-time)	0.00543 Å ² /ps	FAIL	< 0.005 to pass (plateau = stable)
Final MSD	0.411 Å ²	----	
Max displacement	1.075 Å	----	
Coordination (min/mean/max)	5 / 5.0 / 5	PASS	relaxed = 5, tolerance +/-1
Mean bond length (MD vs relaxed)	1.864 Å (relaxed: 1.8	----	
Lattice volume change	+0.00 %	PASS	< +/-5% to pass

Late-time MSD slope = 0.00543 Å²/ps. The slope exceeds the plateau threshold, suggesting mild drift or site-hopping. Final MSD = 0.411 Å²; maximum displacement from starting position = 1.07 Å.

Coordination was exactly 5 throughout the entire trajectory -- a near-perfect result indicating the dopant held its local environment rigidly. Mean nearest-neighbour distance during MD = 1.864 Å.

Volume change = +0.00%. The host lattice is mechanically intact.

VERDICT: MIGRATION (favorable site but dopant relocates during MD)

The formation energy is favorable but Mn migrates away from the Co site during MD. The relaxed structure

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is not the true resting site. Inspect the MSD and trajectory to identify where the dopant ends up.

Mn at the K layer -- MIGRATION / UNFAVORABLE

Charge situation

Mn replaces K with a charge surplus of +2. CHGNet cannot model this compensation (would require Co oxidation or electron addition). The formation energy below is approximate.

Formation Energy (thermodynamic stability)

Formation energy E_f	+3.5509 eV	FAIL	< 1.0 eV to pass
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E_f = +3.551 eV (approximate -- charge surplus not modelled in CHGNet). This value is large and positive, indicating that Mn incorporation is thermodynamically unfavorable under equilibrium conditions at the K site.

Molecular Dynamics Stability (200 C, 25 ps)

MSD slope (late-time)	0.00629 Å²/ps	FAIL	< 0.005 to pass (plateau = stable)
Final MSD	0.165 Å²	----	
Max displacement	0.720 Å	----	
Coordination (min/mean/max)	4 / 5.0 / 5	PASS	relaxed = 5, tolerance +/-1
Mean bond length (MD vs relaxed)	1.871 Å (relaxed: 2.0)	----	
Lattice volume change	+0.00 %	PASS	< +/-5% to pass

Late-time MSD slope = 0.00629 Å²/ps. The slope exceeds the plateau threshold, suggesting mild drift or site-hopping. Final MSD = 0.165 Å²; maximum displacement from starting position = 0.72 Å.

Coordination fluctuated between 4 and 5 (mean 5.0), within the +/-1 tolerance of the relaxed value (5). This is consistent with normal thermal vibration. Mean nearest-neighbour distance during MD = 1.871 Å.

Volume change = +0.00%. The host lattice is mechanically intact.

VERDICT: MIGRATION / UNFAVORABLE

The formation energy is favorable but Mn migrates away from the K site during MD. The relaxed structure is not the true resting site. Inspect the MSD and trajectory to identify where the dopant ends up.

Site Preference Comparison

The Co site is preferred by 6.077 eV over the K site. Formation energy: -2.526 eV (Co) vs +3.551 eV (K).

Mn preferentially occupies the Co layer. The formation energy difference between the two layers is +6.077 eV, strongly favouring the Co site. MD confirms this preference: on the Co site the dopant MSD plateau slope is 0.00543 Å²/ps (stable), compared to 0.00629 Å²/ps on the K site. The Co-site result is isovalent (no charge mismatch), so the formation energy is fully reliable. The K-site E_f is approximate due to a charge surplus of +2 that cannot be modelled in CHGNet.

Caveats and Limitations

? CHGNet is charge-neutral.

The potential does not model oxidation states, polaron formation, or electron/hole localisation.

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Formation energies for aliovalent dopants (charge mismatch $\neq 0$) are approximate even when structural compensation defects are included.

? No Freysoldt/Kumagai correction.

Electrostatic finite-size corrections for charged defects require the host dielectric tensor and the DFT electrostatic potential, neither of which is available from a neural-network potential. For publication-quality E_f of charged defects, follow up with DFT using doped or pymatgen-analysis-defects.

? Metal-rich reference state.

Chemical potentials are computed from elemental ground-state structures. Synthesis under oxide-rich or other conditions shifts all E_f values by additive constants; the relative ordering of sites is unaffected.

? MD timescales are short (25 ps default).

This is sufficient to assess local site stability and rapid migration, but cannot resolve slow diffusion mechanisms. A non-plateauing MSD confirms instability; a plateau does not guarantee stability on longer timescales.

? Energy rankings are more trustworthy than absolute values.

CHGNet is a universal MLIP trained on r2SCAN DFT. Formation-energy errors for transition-metal oxides are typically 50-150 meV/atom. Use relative orderings (which site is preferred?) as the primary result.